Name _____

Choosing Measures and Comparing Distributions KEY

Which measures should you use? It depends on the shape of the data.

Shape of the data	Center	Spread
Symmetric, no outliers	mean	standard deviation
Skewed, or has outliers	median	IQR

Why? An outlier pulls the mean and the SD. It does not pull the median and the IQR much.

Example: 2, 3, 3, 4, 5, 40

	Mean	SD	Median	IQR
With 40	9.5	15.0	3.5	2
Without 40	3.4	1.1	3	2

One value moved the mean by 6.1 and the SD by about 14. The median moved 0.5. The IQR did not move.

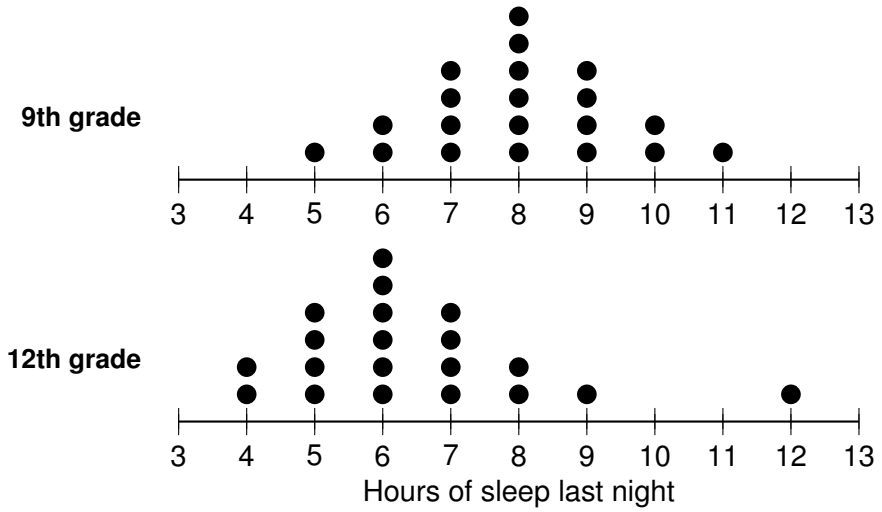
Comparing two distributions. Talk about all four of these. Use numbers and the context.

Shape: symmetric or skewed?

Center: which group is higher? (median or mean)

Spread: which group is more spread out? (IQR or SD)

Outliers: any values far from the rest?



Shape	9th grade is symmetric. 12th grade is skewed right.
Center	The median for 9th grade (8 hours) is greater than for 12th grade (6 hours). 9th graders usually sleep about 2 hours more.
Spread	Both groups have an IQR of 2 hours. The spreads are about the same.
Outliers	One 12th grader slept 12 hours. That is an outlier (the upper fence is 10).

12th grade has an outlier, so we compare with the median and the IQR.

Work with a partner. Compare answers when you finish each problem.

1. For each data set, should you use the **mean and SD** or the **median and IQR**? Explain.

   (a) The heights of 200 adult women. The histogram is symmetric.
   Mean and SD. The data is symmetric with no outliers.

   (b) The price of every house sold in a city. The histogram is skewed right.
   Median and IQR. The data is skewed.

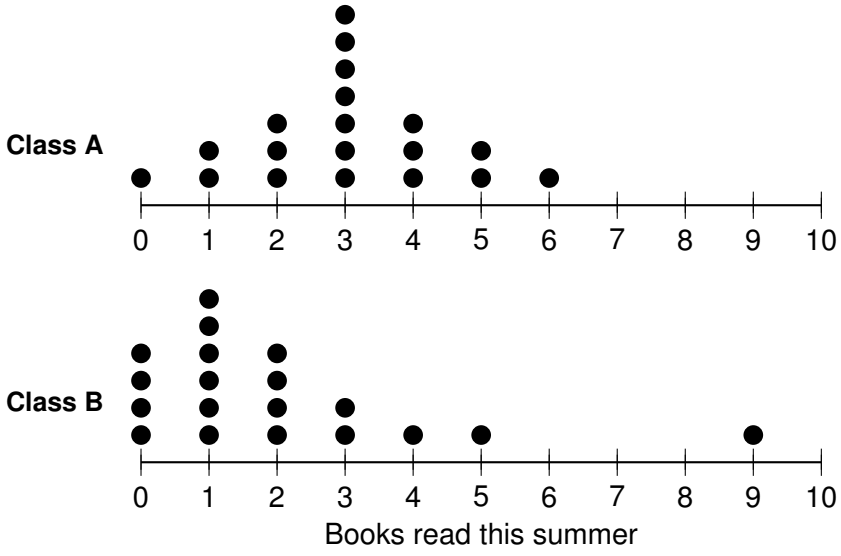
   (c) Quiz scores: 7, 8, 8, 9, 9, 10, 2
   Median and IQR. 2 is an outlier.

2. Look at the example on page 1: 2, 3, 3, 4, 5, 40.

   (a) Which pair describes a typical value better: mean 9.5 and SD 15.0, or median 3.5 and IQR 2? Explain.
   Median 3.5 and IQR 2. Five of the six values are from 2 to 5. The mean of 9.5 is bigger than all of them except 40.

   (b) A student says, "The SD is 15, so the values are very spread out." Is that a fair description? Explain.
   No. Five values are packed from 2 to 5. One outlier makes the SD large.

3. These dot plots show how many books students in two classes read this summer.



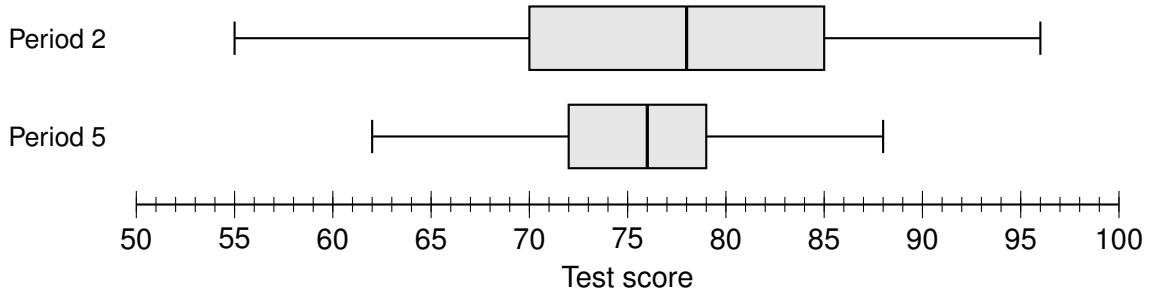
   (a) Describe the shape of each class.
   Class A is symmetric. Class B is skewed right.

   (b) Class A has median 3 and IQR 2. Class B has median 1 and IQR 2. Compare the centers.
   The median for Class A (3 books) is greater than for Class B (1 book). Class A usually read 2 more books.

(c) Compare the spreads.
Both IQRs are 2 books. The spreads are about the same.

(d) Does either class have an outlier? Explain.
Yes. One student in Class B read 9 books. The upper fence for B is $3 + 1.5 \times 2 = 6$.

4. These box plots show test scores in two class periods.



(a) Compare the centers. Use numbers.
The medians are close: Period 2 is 78 and Period 5 is 76. Period 2 is a little higher.

(b) Compare the spreads. Use numbers.
Period 2 IQR = 85 − 70 = 15. Period 5 IQR = 79 − 72 = 7. Period 2 is much more spread out.

(c) Which period's scores are more consistent? Explain.
Period 5. Its IQR and its range are both smaller.

5. Two soccer teams measured the heights of their players. Both histograms are symmetric.

	Mean	SD
Team X	70 inches	2 inches
Team Y	72 inches	5 inches

(a) Which team is taller on average? Team Y, by 2 inches.

(b) Which team has heights that are more spread out? Team Y. Its SD is larger.

(c) Write one sentence that compares the two teams. Use both the center and the spread.
Team Y is taller on average (72 vs 70 inches), and its heights are more spread out (SD 5 vs 2 inches).